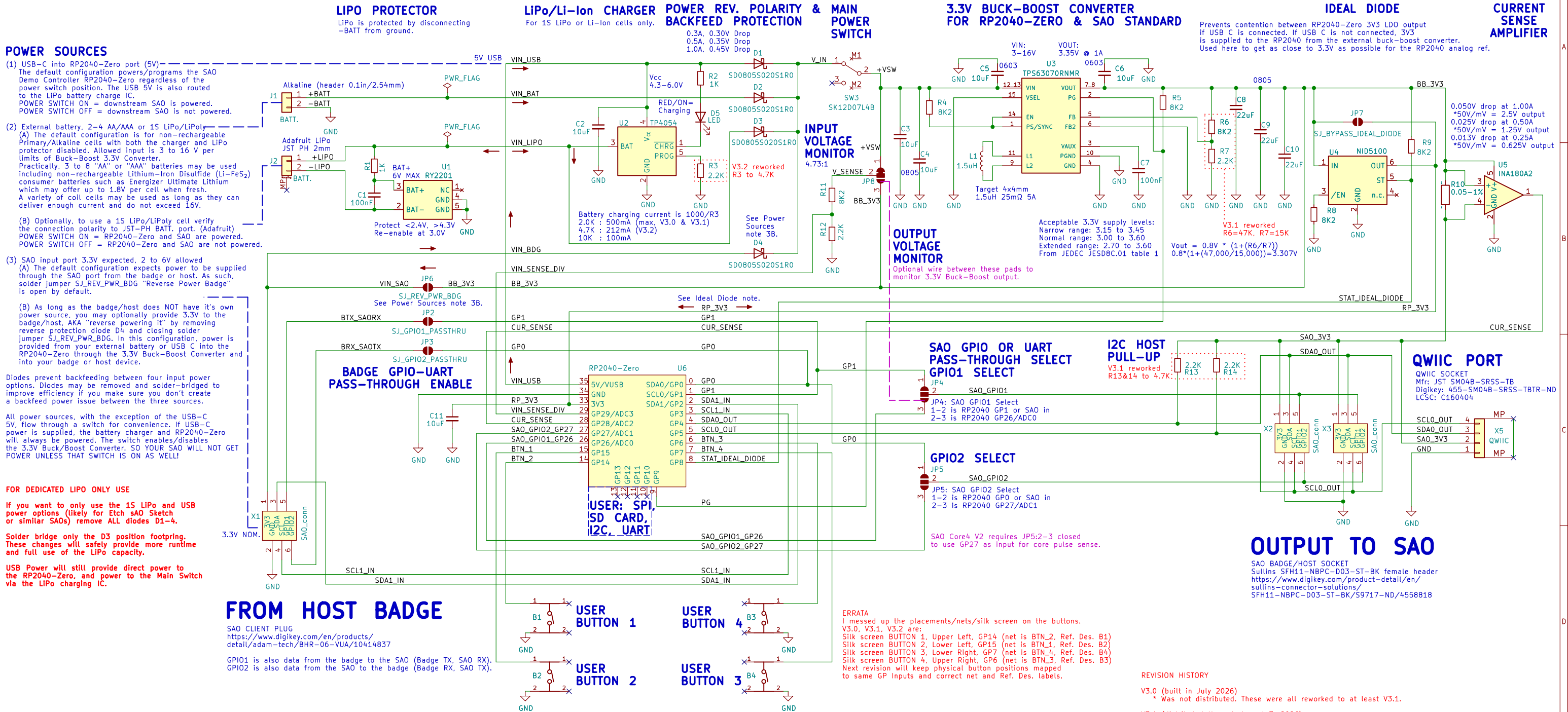


SAO DEMO CONTROLLER V3.0 to 3.2 – A COMPLETE SAO DEVELOPMENT HOST WITH MULTIPLE POWER OPTIONS



OUTPUT TO SAO

SAO BADGE/HOST SOCKET
Sullins SFH11-NBPC-D03-ST-BK female header
<https://www.digikey.com/product-detail/en/sullins-connector-solutions/SFH11-NBPC-D03-ST-BK/S9717-ND/4558818>

REVISION HISTORY

V3.0 (built in July 2026)

- * Was not distributed. These were all reworked to at least V3.1.

V3.1 (distributed through August 3, 2026)

- * Buck-Boost output voltage setting corrected to 3.3V by changing R6/R7 from 8.2K/2.2K to 47K/15K.
- * I2C Pull-ups R13&14 reworked from 2.2K to 4.7K for more reliably I2C operation.
- * Charging circuit has 2.2K current program resistor, 545 mA. This is a little high for the included LiPo pack. Suggest replacing that resistor with 4.7K for 212mA charge current limit.
- * May or may not have V3.1 label added to the board on the top of the USB-C connector.

V3.2 (started distribution August 15, 2026)

- * Reduced LiPo charging current, reworked R3 from 2.2K (454 mA) to 4.7K (212mA)
- * Add V3.2 label.

All non-polarized capacitors are X7R or X5R ceramic unless otherwise noted.

Design by Andy Geppert • www.MachineIdeas.com

Sheet: /

File: SAO_Demo_Controller.kicad_sch

Title: SAO Demo Controller, Micro Badge Development Board

Size: B Date: 2026-08-26

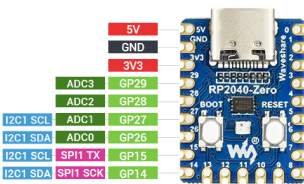
Rev: 3.0, 3.1, 3.2

KiCad E.D.A. 10.0.3

Id: 1/1

MICROCONTROLLER: RP2040-ZERO

DEMO CONTROLLER FUNCTIONS:
INPUT 3-5V from Badge or USB-C
GROUND
3V3 @ ~200mA regulated from ZERO
VIN/2 input voltage measurement
SAO output current measurement



DEMO CONTROLLER FUNCTIONS:

SAO GPIO2 pass-through or Badge RX / SAO TX
SAO GPIO1 pass-through or Badge TX / SAO RX
SDA1 IN from Badge
SCL1 IN from Badge
SDA0 OUT to SAO
SCL0 OUT to SAO
USER BUTTON 3
USER BUTTON 4
USER 5100 STATUS/UART
USER GPIO/IO/UART
USER SPI_CLK/I2C1_SDA/UART
USER SPI_TX_OUT/I2C1_SCL/UART
USER SPI_RX_IN/I2C0_SDA/UART
USER SPI_CS_N/I2C0_SCL/UART

RP2040-Zero
See <https://www.waveshare.com/wiki/RP2040-Zero>
for more information and more GPIO on bottom pads.
Onboard buttons not user accessible.
GPIO16 is NeoPixel onboard.

COMMON I2C ADDRESSES

OLED (SSD1306) 128x32 0x3C
OLED (SSD1306) 128x64 0x3D
OLED (SSD1327) 128x128 0x3C

SAO STANDARDS

PRIMARY: <https://docs.google.com/document/d/1EJqyvkLMAPsQ9VWF5A4elWoi0qMIKyR5Giw5rqRmtmM/edit?usp=sharing>
SUPPLEMENT: <https://hackaday.io/project/175182-simple-add-ons-sao>

QWIC/STEMMA RESOURCES

<https://learn.adafruit.com/introducing-adafruit-stemma-qt/what-is-stemma>